

ChatGPT's take on the prmtop file, which Travis Hughes reviewed and edited/added a few things to:

I parsed `abl_ligand.prmtop` as an AMBER parameter-topology file. In AMBER, a `prmtop` stores the system's atom/residue topology and force-field parameters, organized into `%FLAG` sections such as `ATOM_NAME`, `CHARGE`, `MASS`, `BONDS_*`, `ANGLES_*`, `DIHEDRALS_*`, and `RESIDUE_*`. It does **not** store the atomic coordinates needed to run dynamics; those are normally in an `inpcrd`, `rst7`, NetCDF trajectory, etc. (archive.ambermd.org)

High-level summary

Item	Value
File title	<code>default_name</code>
Version/date stamp	V0001.000, 06/08/26 16:55:03
Total atoms	51,353
Total residues (protein, which is made of many amino acid residues, + number of water, ions, and ligand molecules)	12,052
Total molecules (i.e., water, protein, ions, ligand)	11,786
Net charge	approximately 0 ; computed as about -0.00094 e, due to rounding
Total mass	244,340.44 amu
Periodic box (see	yes
Box type	IFBOX = 2, consistent with a non-orthogonal/truncated-octahedral style periodic cell
Box dimensions	angle 109.471219°; lengths 83.46456 Å × 83.46456 Å × 83.46456 Å
Polarizable model? (i.e., can charges move?)	no, IPOL = 0
Implicit-solvent radii present?	yes, modified Bondi radii (<code>mbondi</code>)
CMAP terms present? (CMAP adjusts dihedral energies for each residue to better fit quantum calculations)	yes, 265 CMAP terms using 14 CMAP grids

System composition

The topology contains a protein, one ligand, ions, and explicit solvent.

Component	Residues/molecules	Atoms	Approx. net charge	Mass
Protein	267 residues / 1 molecule	4,315	-9.000 e	30,985.688 amu
Ligand <code>LIG</code>	1 residue / 1 molecule	67	+1.999 e	481.598 amu

Component	Residues/molecules	Atoms	Approx. net charge	Mass
Sodium Na ⁺	31 ions	31	+31.000 e	712.690 amu
Chloride Cl ⁻	24 ions	24	-24.000 e	850.800 amu
Water WAT	11,729 molecules	46,916	approximately 0 e	211,309.664 amu

The charges add up to essentially neutral: protein -9, ligand about +2, net ions +7, and water neutral.

See <https://ambermd.org/tutorials/basic/tutorial8/index.php> to learn how to add ions to a specific molar concentration.

I have posted a molarity.perl script that automatically calculates the number of ions needed for a desired/input molar concentration and a given/input box dimensions obtained from the last line of the coordinate (.inpcrd) file [83.4645600 83.4645600 83.4645600 109.4712190 109.4712190 109.4712190]

Residue and atom ordering

The residue order is:

Residue range	Atom range	Contents
residues 1–267	atoms 1–4315	protein
residue 268	atoms 4316–4382	ligand LIG
residues 269–299	atoms 4383–4413	31 sodium ions
residues 300–323	atoms 4414–4437	24 chloride ions
residues 324–12052	atoms 4438–51353	11,729 waters

Protein contents

The protein has **267 residues**. Standard residue counts are:

Residue Count	
GLU	28
LEU	25
LYS	19
ALA	19
VAL	18
TYR	15
THR	15
GLY	15
MET	12

Residue Count

PRO	11
ILE	11
ASP	11
ARG	11
SER	11
PHE	11
ASN	9
GLN	8
TRP	7
HIE	6
CYS	5

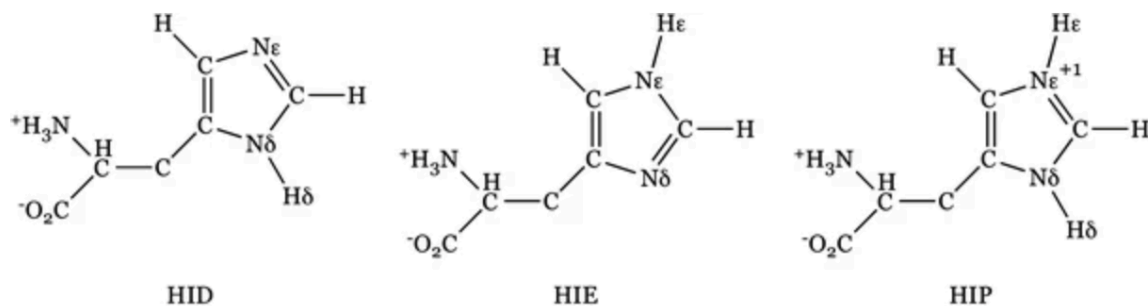
A few topology-specific observations:

The first residue is an N-terminal ASP with atoms N H1 H2 H3, indicating a protonated N-terminus. Its net charge is approximately neutral because the ASP side-chain -1 is offset by the terminal +1. (it is normal for the N-terminal amino acid to have a protonated/+1 carboxyl group at physiologic pH)

The last protein residue is a C-terminal GLU with OXT; its net charge is -2, meaning a normal GLU -1 side chain plus a deprotonated C-terminus. (it is normal for the c-terminal amino acid to have a deprotonated/-1 carboxyl group at physiologic pH)

All histidines are labeled HIE, meaning epsilon-protonated histidine in AMBER naming. I did not see HID or HIP. HID and HIP are alternative protonation states of histidine. HIE and HID are the most common in proteins at physiologic pH.

Fig. 1



Ligand LIG

The ligand is residue **268**, atoms **4316–4382**.

Ligand property	Value
Residue name	LIG
Atom count	67
Approx. formula from atom records	C28 H31 N7 O1
Net charge	+1.998999 e , effectively +2
Mass	481.598 amu
Internal bonds	71
Internal angles	123
Internal dihedrals	215
Covalent link to protein?	none found in the bond list

So this is a noncovalently bound ligand, not a covalent inhibitor or modified residue in this topology.

The ligand uses lower-case GAFF/GAFF-like atom types such as *ca*, *c3*, *c6*, *nx*, *nb*, *hn*, *hx*, *ha*, etc. Its partial charges are atom-specific and sum to about +2.

Water model

Each water residue has **4 atoms**:

Atom	AMBER type	Element/role	Charge
O	OW	oxygen	0.000000 e
H1	HW	hydrogen	+0.679142 e
H2	HW	hydrogen	+0.679142 e
EPW	EP	massless extra point	-1.358284 e

This is a four-site explicit water model with a massless extra point. The file contains **11,729 extra points**, matching the number of waters. The `NUMEXTRA = 11729` pointer reflects these extra/virtual sites.

The water bonding section includes O–H, H–H, and O–EPW topology terms. No water angle terms are stored; the geometry is represented through bond-style constraints/terms.

Atom identity sections

The main atom-level sections each contain **51,353 entries**:

Section	Meaning
ATOM_NAME	atom names such as N, CA, O, C1, EPW
CHARGE	AMBER-scaled partial charges; divide by about 18.2223 to get elementary charge
ATOMIC_NUMBER	element number; 0 marks extra points
MASS	atomic masses
ATOM_TYPE_INDEX	numeric nonbonded/Lennard-Jones type index
AMBER_ATOM_TYPE	text force-field atom type
NUMBER_EXCLUDED_ATOMS and EXCLUDED_ATOMS_LIST	nonbonded exclusions, mainly bonded 1–2/1–3 and special-case exclusions

Element counts in the full system are:

Element/role	Count
H	25,625
O	12,134
Extra point EP	11,729
C	1,431
N	362
Na	31
Cl	24
S	17

Force-field parameter sections

The topology stores both parameter tables and the connectivity that references those tables.

Bonded terms

Quantity	Count
Bond parameter types	93
Angle parameter types	204
Dihedral parameter types	207
Bonds involving hydrogen	37,354
Bonds without hydrogen	14,004
Total bonds	51,358

Quantity	Count
Angles involving hydrogen	4,941
Angles without hydrogen	3,079
Total angles	8,020
Dihedrals involving hydrogen	9,937
Dihedrals without hydrogen	9,562
Total dihedrals	19,499

Connectivity is stored in AMBER's compact integer format. For `BONDS_*`, each record is three integers: atom 1, atom 2, and bond-parameter index. The atom numbers in these arrays are stored as $3 \times \text{zero-based_atom_index}$, so software converts them back to normal atom serials. Angles and dihedrals use similar compact arrays.

Bonded topology by component:

Component	Bonds	Angles	Dihedrals
Protein	4,371	7,897	19,284
Ligand	71	123	215
Water	46,916	0	0
Ions	0	0	0

Nonbonded terms

The file has:

Section	Count
<code>NTYPES</code> Lennard-Jones type indices	30
<code>NONBONDED_PARM_INDEX</code> entries	$900 = 30 \times 30$
<code>LENNARD_JONES_ACOEF</code> entries	465
<code>LENNARD_JONES_BCOEF</code> entries	465

AMBER stores Lennard-Jones pair interactions as $A/r^{12} - B/r^6$. The `NONBONDED_PARM_INDEX` matrix maps pairs of atom-type indices to entries in the A and B coefficient arrays.

Some important LJ type-index assignments in this file:

Type index	Main atom type(s)	Count	Notes
13	HW, HO	23,499	water/protein hydrogens with zero LJ parameters

Type index	Main atom type(s)	Count	Notes
29	OW	11,729	water oxygen LJ site
30	EP	11,729	water extra point, zero mass and zero LJ
27	Na+	31	sodium ion
28	Cl-	24	chloride ion
16–26	ligand-like lower-case types	67 total	ligand-specific GAFF-like types

CMAP correction sections

The file contains CMAP sections:

CMAP item	Value
CMAP_COUNT	265 CMAP terms, 14 maps
CMAP_RESOLUTION	14 grids, each 24×24
CMAP_PARAMETER_01–CMAP_PARAMETER_14	each has 576 values
CMAP_INDEX	265 records $\times$ 6 integers

These are backbone correction maps. The 265 CMAP terms correspond to internal protein residues; the first and last residues do not get the same full five-atom backbone map. This is consistent with a modern AMBER protein force field using CMAP-style backbone corrections.

Solvent and periodic boundary sections

SOLVENT_POINTERS = [323, 11786, 58] means:

Value	Interpretation
323	last non-solvent residue index
11,786	total molecules
58	first solvent molecule index

This agrees with the residue/molecule ordering: molecule 1 is protein, molecule 2 is ligand, molecules 3–57 are ions, and molecule 58 onward are waters.

BOX_DIMENSIONS = [109.471219, 83.46456, 83.46456, 83.46456]. In AMBER topology convention, this stores the box angle followed by box lengths. The equal lengths and 109.471219° angle indicate a truncated-octahedral-style periodic system.

GB/implicit-solvent radii sections

Even though the system has explicit water, the file also includes `RADII` and `SCREEN` arrays. These are commonly present in AMBER topologies and are used by generalized Born / implicit solvent calculations or tools that need intrinsic radii.

Section	Meaning
<code>RADIUS_SET</code>	modified Bondi radii (mbondi)
<code>RADII</code>	per-atom intrinsic radii
<code>SCREEN</code>	per-atom screening parameters

Typical values in this file include carbon radius 1.70 Å, nitrogen 1.55 Å, oxygen 1.50 Å, sulfur 1.80 Å, and many hydrogens at either 1.30 Å or 0.80 Å.

Practical interpretation

This topology represents a **neutral, explicitly solvated ABL–ligand system** with:

A 267-residue protein, one noncovalent +2 ligand, 31 sodium ions, 24 chloride ions, and 11,729 four-site waters. The protein appears to use a CMAP-enabled AMBER force field. The solvent is a four-site water model with one massless extra point per water. The topology is periodic, with a truncated-octahedral-style box around 83.46 Å.

The most important caveat is that this file is only the **topology and parameters**. To actually run or inspect the structure, you also need the matching coordinate/restart file, such as `abl_ligand.inpcrd`, `abl_ligand.rst7`, or an equivalent coordinate file.